Introduction to Abstract Algebra

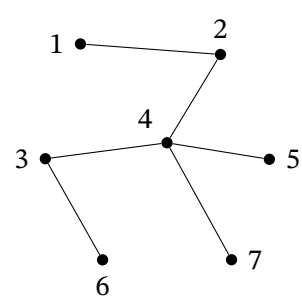
Practice Midterm 1, March 23, 2026

1. Show that the binary operation
$$a * b := \frac{a+b}{ab}$$
is not a group operation on $\mathbb{R}_{>0}$.

2. Show that the binary operation
$$a * b := \frac{a+b}{1+ab}$$
defines a group on the open interval $G := (-1, 1)$. Compute explicitly the identity element and the inverse a^{-1} of any $a \in G$.

3. Use the function
$$f(x) = \frac{1+x}{1-x}$$
to show that the group $G := (-1, 1)$ with the binary operation of the previous exercise is isomorphic to the group $(\mathbb{R}_{>0}, \cdot)$.

4. Consider the subset
$$T := \{(12), (24), (34), (36), (45), (47)\} \subseteq S_7,$$
so that each element of T is a transposition. We can use the set T to draw a graph by connecting a and b with an edge if $(ab) \in T$. This way, we obtain:



   a) In the above graph, there is a path from 2 to 6. Use this path to show that the transposition (26) is an element of the subgroup $\langle T \rangle$ generated by T.
   b) Deduce that every trasposition (ab) lies in the subgroup generated by T, and conclude that $\langle T \rangle = S_7$.

Hints

1. Suppose that e is an identity element. Derive a contradiction.

2. Show first that the operation is well-defined, meaning that we have

$$-1 < \frac{a+b}{1+ab} < 1$$

   for every $a, b \in (-1, 1)$. Then prove the group axioms. For a clear proof, state which axiom you are about to prove before proving it.

3. Show that this function is a bijection $(-1, 1) \to (0, \infty)$ and preserves the group operation.

4. In part (a), use that $(ab)(bc)(ab) = (ac)$. In part (b) observe that there is a path between any two points, so part (a) can be generalized to conclude that every transposition is in $\langle T \rangle$. Since the transpositions generate the symmetric group, it follows that T generates the symmetric group.

   General observation: If a set T of transpositions determines a connected graph on the set $\{1, \dots, n\}$, then T generates S_n.

Answers

Note that it is important to clearly structure your proofs. In the answers below, I have emphasized some the sentences and key words that indicate the structure of the proof.

1. *We will prove that there is no identity element, using a proof by contradiction.* Suppose that e is an identity element. Then we have
$$\frac{1+e}{1e} = 1.$$
This shows that $e = e + 1$, a contradiction.

2. *We need to prove closure and the group laws.*

   Closure. *For closure, suppose that $a, b \in (-1, 1)$.* In order to show that
$$-1 < \frac{a+b}{1+ab} < 1$$
we have to show that $-(1+ab) < a+b$ and $a+b < 1+ab$. For the first strict inequality, note that $-(1+ab) < a+b$ if and only if $0 < 1+ab+a+b$. Since $a, b > -1$, it follows that
$$1+ab+a+b = (1+a)(1+b) > 0.$$
For the second strict inequality, note that $a+b < 1+ab$ if and only if $1+ab-a-b > 0$. Since $a, b < 1$, it follows that
$$1+ab-a-b = (1-a)(1-b) > 0.$$

   Group laws. *Next, we need to prove the group laws.* To see that the operation $a * b$ is *associative*, we compute
$$(a*b)*c = \frac{\frac{a+b}{1+ab}+c}{1+\frac{a+b}{1+ab}c} = \frac{a+b+c+abc}{1+ab+ac+bc}.$$
Similarly, we have
$$a*(b*c) = \frac{a+\frac{b+c}{1+bc}}{1+a\frac{b+c}{1+bc}} = \frac{a+b+c+abc}{1+ab+ac+bc},$$
so we see that $(a*b)*c = a*(b*c)$.

   We note that the operation $a*b$ is commutative, so for *identity laws* and *inverse laws* we have to prove only one case each. The identity element is 0, since
$$\frac{a+0}{1+a0} = a.$$
The inverse of a is $-a$, since
$$\frac{a-a}{1-a^2} = 0.$$

3. *First, we need to show that the function $f : (-1,1) \to (0,\infty)$ is a bijection.* To prove that f is *injective*, suppose that $f(a) = f(b)$. Then we have $(1+a)(1-b) = (1+b)(1-a)$, which gives

$$1 + a - b - ab = 1 - a + b - ab.$$

This equation holds if and only if $a - b = b - a$, which holds if and only if $2a = 2b$, which holds if and only if $a = b$. To prove that f is *surjective*, let z be a positive integer. If $f(a) = x$, then we have

$$x(1-a) = 1 + a,$$

which is equivalent to $x - 1 = a(1+x)$. This shows that

$$f\left(\tfrac{x-1}{1+x}\right) = x.$$

This completes the proof that f is a bijection.

Next, we need to show that f preserves the group operation:

$$f(a * b) = \frac{1 + \frac{a+b}{1+ab}}{1 - \frac{a+b}{1+ab}} = \frac{1 + ab + a + b}{1 + ab - a - b} = \frac{(1+a)(1+b)}{(1-a)(1-b)} = f(a)f(b).$$

4. a) Recall that $(ab)(bc)(ab) = (ac)$. Using this equation, we find that

$$(24)(43)(36)(43)(24) = (24)(46)(24) = (26).$$

b) If we have a path $x_0 - x_1 - \cdots - x_n$ in the graph, then we can write

$$(x_0 x_n) = (x_0 x_1) \cdots (x_{n-2} x_{n-1})(x_{n-1} x_n)(x_{n-2} x_{n-1}) \cdots (x_0 x_1).$$

This follows from repeatedly applying the identity $(ab)(bc)(ab) = (ac)$. Since the graph of T is connected, there is a path between any two vertices. Thus we conclude that $\langle T \rangle$ contains all the transpositions, so we have $\langle T \rangle = S_7$.